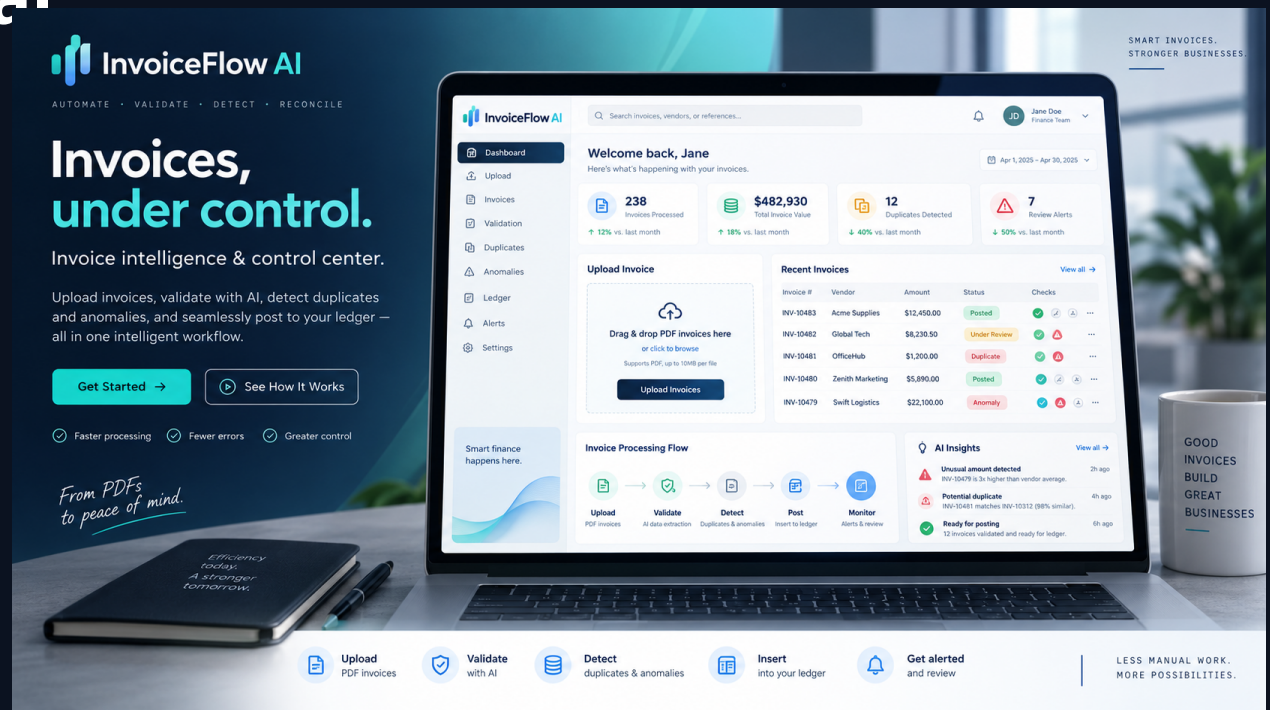


INVOICE AUTOMATION + OPERATIONAL INTELLIGENCE

From PDF invoices to controlled financial operations.

InvoiceFlow AI is an automated invoice-processing MVP that extracts, validates, reviews, stores, alerts, and reports - turning manual invoice handling into a structured operational workflow.



Python

Streamlit

n8n Cloud

Telegram

Power BI

Conceptual product visualization - presentation mockup, not a literal production screenshot.

The business problem

Invoice handling is often fragmented across PDFs, spreadsheets, email, and manual review. The result is slower processing, duplicated effort, inconsistent controls, and limited visibility into spend and exceptions.

Before automation

- **Manual extraction**

Invoice fields are copied into spreadsheets or accounting tools by hand.

- **Scattered validation**

Duplicate, VAT, date, and total checks are easy to miss or apply inconsistently.

- **Delayed visibility**

Supplier, overdue, and exception insights arrive after the operational work is already done.

With InvoiceFlow AI

- **Structured extraction**

Invoice metadata and financial values are parsed from uploaded PDFs.

- **Operational controls**

Validation, overdue, duplicate, and supplier anomaly checks run automatically.

- **Centralized follow-up**

Master Ledger, alerts, and dashboards provide one operational view.

SYNTHETIC INVOICES

250

PDF LAYOUTS

3

CURRENCIES

3

WORKFLOW OUTCOMES

3

A controlled path from upload to action

The private implementation uses a multi-step automation pipeline for parsing, validation, review routing, ledger updates, alerts, and reporting.



Conceptual workflow visualization. Production node code and implementation details are intentionally withheld from the public repository.

The system does more than read invoices.

Each invoice is converted into a set of financial and operational checks before the workflow decides whether it can proceed normally or needs attention.

Required fields

Completeness

Core invoice, supplier, date and financial fields.

Financial validation

VAT & totals

Arithmetic and currency-aware VAT rules.

Operational control

Overdue

Paid invoices are excluded from overdue status.

Duplicate protection

Duplicate

Invoice number + supplier blocks repeat insertion.

Supplier intelligence

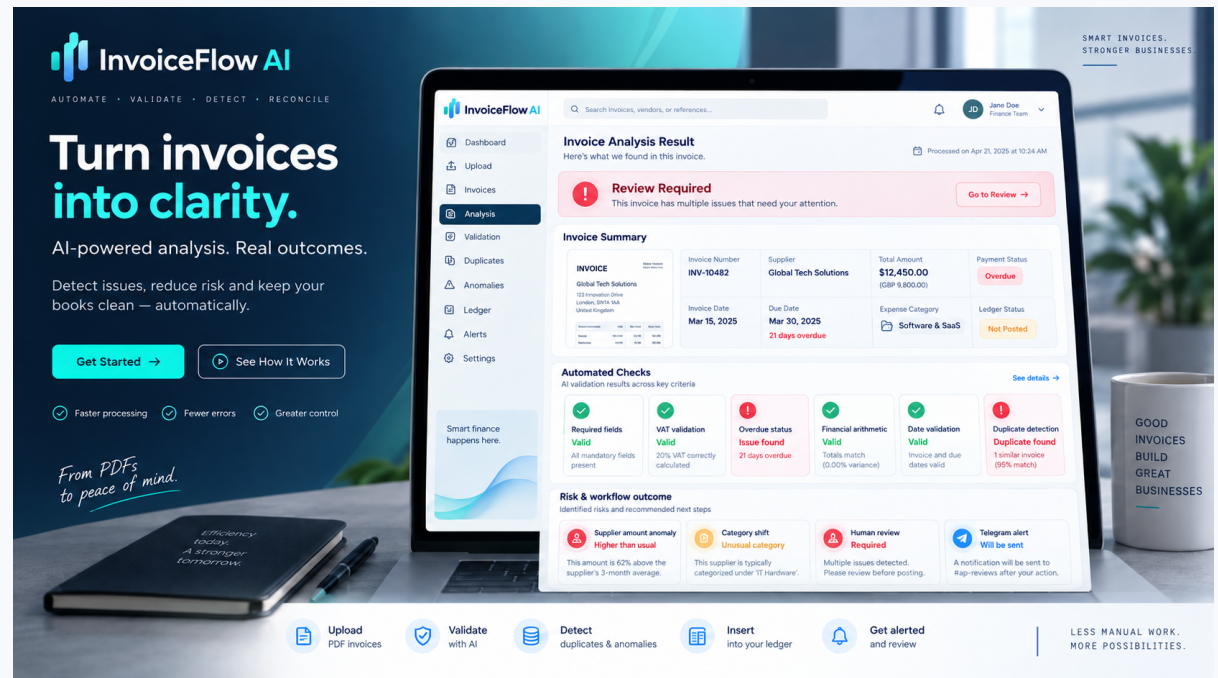
Amount anomaly

Same-supplier, same-currency historical comparison.

Behavior pattern

Category shift

Unexpected supplier category patterns can trigger review.



Conceptual product visualization with illustrative values and UI details.

Operational insights from the Master Ledger

The dashboard is designed around a dedicated data API backed by the complete n8n Master Ledger, so reporting is not limited to one local browser session.



What the dashboard monitors

- **Processed & ledger invoices**
Operational throughput and stored records.
- **Review & overdue**
Exceptions and open invoices needing attention.
- **Supplier & category views**
Spend concentration and operational patterns.
- **Currency-aware totals**
AED, SAR and USD remain separated.

Validated production branches

- Clean invoice
- Review required
- Duplicate
- Dashboard API

Public release boundary:

Parser code, workflow exports, production source, credentials, datasets, Power BI files, and operational endpoints remain private.